

Zero-shot Avian Species Detection from Unlabelled Field Audio Data

Gayathri Singaram¹, Lakshmi Babu Saheer², Dena Jane Clink³, Roeun Sala⁴, Moeurk Hong⁵,
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^{1 2} Anglia Ruskin University, United Kingdom, ³ K. Lisa Yang Center for Conservation Bioacoustics, Cornell Lab of Ornithology, Cornell University, USA, ^{4 5 6} Jahoo, Angdoung Kraleung Village, Sen Monorom Orang, MondulKiri, Mondulkiri Province, Cambodia.

gs886@student.aru.ac.uk, lakshmi.babu-saheer@aru.ac.uk, djc426@cornell.edu,
research@gibbon.life, booking@gibbon.life, helene.biot@worldhope.org

Abstract

The automated identification of bird species is a highly complex task, as environmental noise and variations in bird sounds are present in field recordings. This study focuses on identifying 7 bird species from Cambodian field recordings. Due to the lack of labelled in-domain data, zero-shot learning approach is used. Custom & pre-trained models: BirdNet, YAMNet, VGGish, InceptionV3, ResNet50, Xception, RNN+LSTM & CNN are trained on Xeno-Canto & e-bird datasets as audio (MFCC and Chroma features) & spectrogram-based image tasks. This research reveals that MFCC features boost the accuracy of CNN in audio tasks, highlighting its adaptability. Custom models outperformed pre-trained models in audio feature tasks with 99% accuracy, while pre-trained models excel in spectrogram tasks with 92% accuracy on the validation sets. All models had low performance on test data, which could be attributed to the bias towards oriental pied hornbill & great hornbill species due to data imbalance.

Index Terms: Bioacoustics, Signal Processing, Bird Sound Identification, Deep Learning, CNN, RNN, LSTM, BirdNet, MFCC, Chroma features.

1. Introduction

Avian acoustics research, essential for understanding bird behavior and conservation, has progressed from traditional, labor-intensive sound detection methods which involves manual validation to advanced automatic recognition using deep learning and advanced signal processing, enhancing accuracy and efficiency. Recent advancements in bird vocalization analysis have highlighted semi-supervised learning and deep clustering as emerging yet evolving techniques for identifying bird species in field recordings. A recent study explores zero-shot learning (ZSL) for bird species identification using field guide illustrations, [1] introduces a contrastive encoding method and Prototype Alignment to map illustrations and photographs to a shared space. Testing on the iNaturalist2021 dataset, they achieve 12% top-1 and 38% top-10 accuracy for unseen species, showcasing the effectiveness of illustrations in ZSL, another review [2] emphasizes the importance of accurate pre-processing like adaptive denoising while exploring the evolution of feature extraction methods from manual to automated approaches (like CRNN and WaveNet along with methods such as feature fusion and network architecture search to enhance model performance. The review suggests combining visual (spectrogram images) and acoustic data to improve bird species identification using generalised models with larger datasets to aid biodiversity conservation.

Right features and feature fusion methods help the classification process. Studies have considered different

feature-based approaches to classify bird sounds like MFF-ScSEnet, a novel method for bird song identification that combines Mel-filters and Sincnet-filters (filters that process raw audio) with the ResNet18 network and a ScSEnet attention module to enhance spectrogram analysis[3]. This approach significantly improves accuracy in acoustic feature extraction but struggles with mixed or limited samples. [4] introduces Multi-scale CNN and Ensemble Multi-Scale CNN models that use the wavelet transform for spectrogram feature generation, outperforming traditional CNNs in bird species recognition by capturing detailed frequency information.

Bird sound detection faces challenges like environmental noise and the complexity of various vocalizations especially in a 24X7 field recording setting. Researchers [5, 6] have highlighted the acoustic diversity and complexity among bird species, noting variations due to geographical differences and recording distances, leading to high intra-species variation, which complicates accurate species characterization. Researchers [7, 8] address the issue of limited and unbalanced training data in bird sound recognition, emphasizing the need for large, representative datasets to prevent model overfitting and to capture species variability, pointing out the critical gap in the availability of verified datasets for classifier training.

Different studies on bird sound recognition have been performed using audio and spectrogram images: [9] presents "Transound", leveraging a vision transformer and MFCC for effective bird mitigation at airports. [10] combined CNNs with a transformer encoder, enhancing bird sound identification by using multiple acoustic features. [11] proposes the SFLN (spectrogram frame linear network) method for classifying bird sounds based on continuous frame sequences and spectrogram analysis, demonstrating superior performance compared to previous methods. [12] introduces a de-noising method combining wavelet packet decomposition with filtering, enhancing noise reduction in bird recordings and suggesting potential broader applications.

In the world of avian bioacoustics, the BirdCLEF challenges have consistently pushed the boundaries of bird call recognition technology with diverse methodologies like multiple data augmentation[13] and ensemble techniques[14] for robustness against environmental noise. [15] addressed challenges of memory management, species variety, and signal-to-noise ratio variance using pre-trained models and data augmentation, with Inception-v3 outperforming ResNet.

While existing studies focus on well-curated labeled data, this research aims to identify seven specific bird species (4 rare and 3 endangered) from unlabelled field recordings in Jahoo, Cambodia. The study introduces a novel deep learning approach, advancing towards self-

supervised learning. Pre-trained models are retrained with standard datasets to ensure comparability with custom models, thus setting new benchmarks in the field. Through extensive exploration of bird calls and songs using audio and spectrogram analysis, the research diverges from traditional methods that rely on segmented or controlled data. Analyzing raw data from complex environments, the study demonstrates the potential of deep learning models to handle the unpredictability of real-world data. By challenging existing methodologies, it establishes new standards for bioacoustic research, which is crucial for effective conservation strategies.

2. Multi-Modal Project Approach

In this project, bird species identification is divided into two separate tasks: Bird call and Bird song identification using two different approaches: audio features and spectrogram images for seven bird species namely, Germain's peacock pheasant, Giant ibis, Great hornbill, Orange necked partridge, Oriental pied hornbill, White-shouldered ibis, White-rumped Shama. These 7 species are expected to be prominent in the recordings as per the field experts.

2.1. Dataset

Testing data for this project was provided by Cornell University in collaboration with Cambodian collaborators, where data was acoustically collected by deploying 10 SwiftOne units, programmed to continuously record with a 30 dB gain and a 32 kHz sample rate where data has been in continuous passive acoustic monitoring since 2022. The dataset comprises extensive field recordings of avian sounds, from which 53 hours of audio data was specifically chosen to align with bird breeding seasons, to ensure the diverse range of bird calls and songs because audio recordings were not categorized in calls and songs. The data was pre-processed to remove the noise and clean the data after acquiring the noise profile of raw data to understand ambient noise characteristics present in the audio files. The appropriate threshold factors were empirically determined for the power spectral density of audio from the range of noise threshold factors between '3.0' to '-12.5dB'. Following the noise reduction, audio files were segmented into 4-second clips, resulting in 38141 samples.

Training data for this project was collected from two sources "Xeno-canto" and "e-bird", for the aforementioned seven target species, including audios and sonograms of bird calls and songs. The collected audio files were examined using Audacity, and segmented into individual calls and songs, and labeled according to the species name. The audio data consists of 1187 calls and 253 songs, and spectrogram image data consists of 302 calls and 338 songs. Preprocessing for the audio task involved converting all samples to the sampling rate of 44KHz and using a single channel, for the image task spectrograms were resized to 224×224 pixels, and images were converted to RGB format. The train and validation data split was in the ratio of 80:20 to compare the model performances.

Two primary features were extracted:

MFCC: Parameters set included 13 MFCC features, with a Fast Fourier Transform window length of 2048 samples and a hop length of 512, resulting in each audio file being characterized by 13 MFCC features.

Chroma Features: Combined Chroma CQT and Chroma CENS features were used to capture pitch and harmony, with 12 features from each type per audio file and

a hop length of 512. This led to each audio file with 24 Chroma features with feature fusion.

2.2. Modelling

Experiments are performed using custom and pre-trained CNNs (for both audio and spectrogram images) and RNN+LSTMs (for the audio task) after empirically optimising the architecture and hyper-parameters in each case.

The **custom CNN** architecture begins with a masking layer, followed by two convolutional layers (using 32 and 64 filters) for feature extraction, batch normalization, and max pooling for stability and dimensionality reduction, and dropout layers to prevent overfitting. It transitions to fully connected layers, ending with output layers for different data classes (6 for calls, 5 for songs), totaling 1,416,710 parameters.

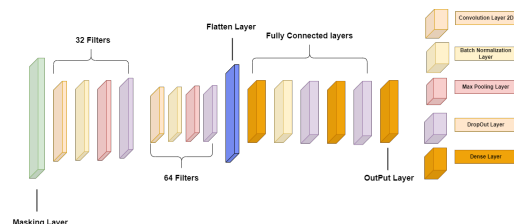


Figure 1: CNN Architecture

The **RNN+LSTM** architecture is designed for recognizing long-term dependencies in sequential data, with LSTM layers arranged to increase (64, 128, 256) and then decrease (128, 64) in units. It includes Dropout layers for preventing overfitting, followed by a flattened layer for transitioning to a dense network structure and ending with dense layers for feature refinement and classification (output of 5 and 6 classes for songs and calls respectively). This architecture boasts over 27 million trainable parameters, indicating its complexity and capability for multi-class classification tasks.

BirdNet analyzer was initially employed to label the test data, the preprocessed test data was fed to the analyzer and it resulted in the prediction of a species count of 5117, in which one of the target species *White Rumped Shama* was identified. Later, BirdNET Analyzer was utilized for

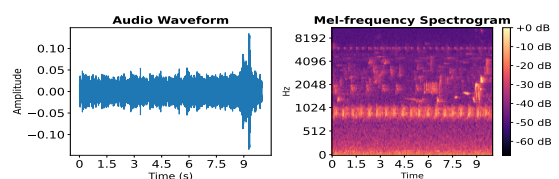


Figure 2: Target Species: White Rumped Shama (visualizing waveform in amplitude and spectrogram in frequency aligns with human perception and facilitates clear understanding)

both training and generating predictions on test data. Some modifications to the original BirdNET architecture like the 'Adam' optimizer were employed with specific optimised learning rates and accuracy metrics.

YAMNet and **VGGish** models involved preprocessing audio files according to model requirements and resampling to model standard rate. Audio for these two models were extracted using YAMNet and VGGish embeddings. Data augmentation and custom layers (neural network with

dense, dropout regularization and softmax) were applied to YAMNet, A Custom CNN classifier was designed and optimised to use the VGGish extracted embeddings.

ResNet-50, Inception V3, and Xception models were adapted by removing their top layers, with Global Average Pooling layer and customized with a dense layer of 1024 neurons using ReLU activation, Inception V3 was enhanced with two dense layers and a 50% dropout rate. Softmax layer finalized the architectures, ensuring the pre-trained weights remained frozen. Adam or RMSprop optimizers alongside categorical cross-entropy loss were selected for model compilation, optimizing performance.

3. Experiments

Models are evaluated using the unseen labelled validation set split out of the training data. Based on the validation performance, one or more models could be used to label the test data using a voting scheme.

3.1. Audio Task

Task	Features	CNN				RNN+LSTM			
		A	P	R	F1	A	P	R	F1
Calls	MFCC	99	99	100	99	88	88	85	86
Calls	Chroma	81	86	85	85	71	69	61	62
Songs	MFCC	94	77	79	76	78	62	66	63
Songs	Chroma	66	44	49	45	70	61	64	61

Table 1: Deep Learning Audio Model Results, A=Accuracy, P=Precision, R=Recall, F1=F1 score

Deep CNN and RNN+LSTM models were trained on MFCC and chroma features separately(table 1). It was revealed that models utilizing MFCC features markedly surpassed those with chroma features in audio classification. Specifically, the CNN model reached a 99% validation accuracy for call classification with high precision and recall, which slightly decreased for songs, in contrast to 94% for calls and 66% for songs with chroma features. Similarly, the RNN+LSTM model exhibited enhanced performance with MFCC, achieving 88% accuracy for calls and 78% for songs, with performance decreasing when trained on chroma. This trend highlights the importance of feature selection, with MFCC features significantly boosting model accuracy and validation accuracy. Germain's peacock pheasant was consistently recognised in both (calls & songs) task(figure 3), along with giant ibis and great hornbill in calls and songs task with very few mis-classifications and higher precision and recall rates, emphasizing role of MFCC's in audio classification.

In the evaluation of pre-trained models for audio classification, BirdNet, even with achieving 99% training accuracy for both calls and songs on the train set, could not generate even a single validation label correctly. YAMNet, using its embeddings, demonstrated robust performance with a 94% accuracy and high precision and recall rates for calls, though it experienced a significant performance drop in songs task(table 2). VGGish, while underperforming on its own, significantly improved when integrated with a CNN-2D model, reaching validation accuracies of 95% for calls and 85% for songs, with good precision and recall scores for both tasks, highlighting the combination of VGGish features and the CNN architecture. It was also noted that the great hornbill was recognised across all pre-trained models, with very few misclassifications.

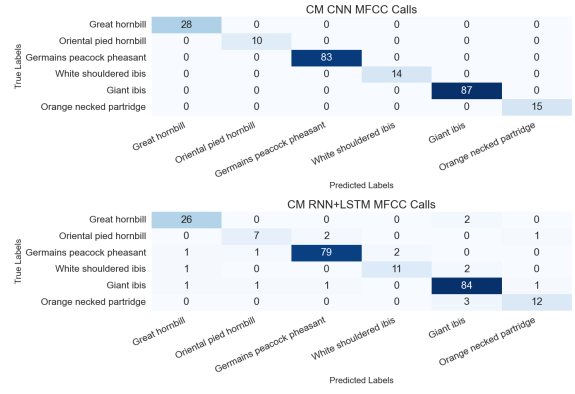


Figure 3: Confusion matrix - Best performing deep learning audio models(CNN & RNN+LSTM in Calls task with MFCC) (Results represent audio task where training set has 6 species).

Models	Calls				Songs			
	Acc	P	R	F1	Acc	P	R	F1
YamNet	94	89	88	88	80	79	68	69
CNN+Vggish	95	84	83	83	85	74	76	72

Table 2: Pretrained Models in Audio Classification, Acc=Accuracy, P=Precision, R=Recall, F1=F1 score

3.2. Spectrogram Image Task

Feature selection in image processing: CNNs learn visual features through multiple layers and filters. Inception V3 targets features at multiple scales with its varied layers in the inception module, while ResNet50 uses residual connections to aid in training deeper networks by overcoming the vanishing gradient issue. Xception improves upon Inception with depthwise separable convolutions for efficient, channel-wise feature learning. In spectrogram-based image classification tasks, 8 models were employed in total for songs and calls.

Model	Calls				Songs			
	Acc	P	R	F1	Acc	P	R	F1
CNN	91	40	41	41	89	88	33	24
InceptionV3	88	72	66	67	92	76	58	49
ResNet50	80	48	51	49	86	98	25	27
Xception	88	71	75	71	91	65	44	61

Table 3: Spectrogram Image Model Results, Acc=Accuracy, P=Precision, R=Recall, F1=F1 score

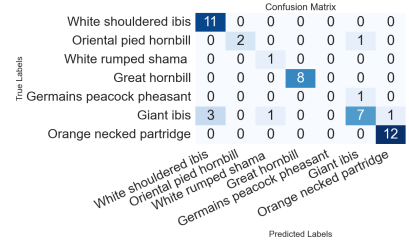


Figure 4: Xception Spectrogram classification in calls (Results represent Spectrogram image task trained with 7 species)

It is observed that CNN models maintained consistency in terms of training accuracy of 87% across both calls and songs tasks, with a slightly higher validation accuracy for calls 91% compared to songs 89% (table 3). This suggests better generalization for calls within the validation dataset. In terms of class identification, the model performed better on calls, correctly identifying 4 out of 7 classes, compared to 2 out of 6 for songs. Precision and recall rates were the lowest.

Model	Dataset	Task	ACC	Reference
Transound	Xeno-Canto	Audio	95	[9]
Inception	Xeno-Canto	Audio	94	[15]
Custom CNN	Xeno-Canto	Audio	99	This Study

Table 4: Comparison of Previous studies with this study on Xeno-canto dataset using MFCC features

In a comparison of state-of-the-art pretrained models, Xception and InceptionV3 demonstrated superior performance on the calls task, with 88% validation accuracy, both identifying 6 out of 7 classes but with lower precision and recall rates. Xception also excelled in the songs task, maintaining 91% validation accuracy, showcasing its strong feature extraction for image data. Conversely, ResNet50 was less effective, with 86% accuracy for songs and 80% for calls, identifying the fewest classes in both tasks. These results highlight the significance of selecting models based on task specificity, with Xception emerging as the top performer (figure 4) for both tasks due to its high accuracy and class identification capabilities.

4. Results & Evaluation

The comparative analysis with previous studies on Xeno-Canto dataset (Table 4) highlights the effectiveness of deep learning models, particularly custom CNN architecture, in leveraging MFCC features for accurate audio classification. It can be observed that the proposed model outperforms state-of-the-art models. This emphasizes the importance of model selection and customization in optimizing performance.

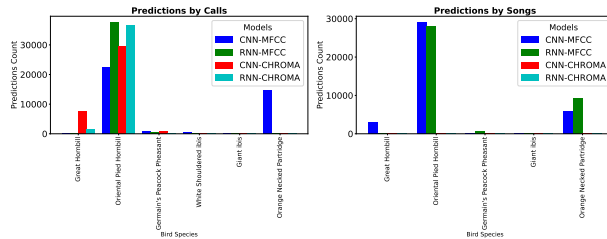


Figure 5: Audio task deep models predictions calls & songs (Results represent audio task where training set has 6 species) (Low representation of species like giant ibis and white-rumped shama in training data significantly impact model predictions)

Post-training, models were tested on unlabeled data to evaluate how many species can be identified from field recordings (Figure 5). Custom deep learning models outperformed pre-trained ones, especially in audio classification. However, biases were noted. For calls, CNN models with MFCC features identified species like the Oriental Pied Hornbill with high confidence, but less represented species were often missed, indicating overfitting. RNN+LSTM models detected a broader range of species but with lower confidence.

For song tasks, pre-trained models like YAMNet predicted a wider spectrum of species but with moderate confidence, needing further refinement. Spectrogram image tasks revealed additional challenges: custom CNN and pre-trained models like InceptionV3 & ResNet50 often identified the Great Hornbill in calls, and heavily favored the Giant Ibis in songs, showing a tendency to overfit. These issues highlight the need for balanced data representation and improved model tuning to ensure equal species recognition.

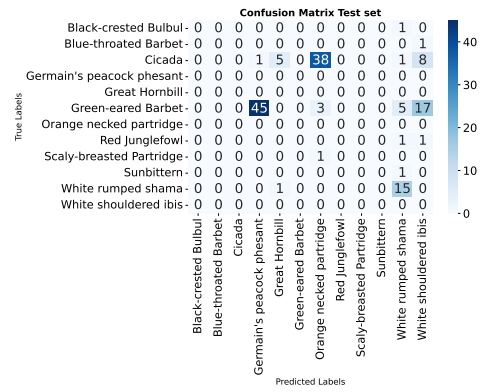


Figure 6: Unseen test set (raw field data) Results

Evaluation of Model Performance on Unlabeled Data The predicted labels must be validated once the species are identified in the field recordings. The evaluation of model performance on unlabeled data was conducted using ensemble voting approach, manual validation, and comparison with a reference model (BirdNet). This multi-step process ensured a comprehensive assessment of the model's accuracy and reliability. To determine species predictions, an ensemble voting method was employed to generate consensus labels for each recording by implemented models. A subset of 50 samples per species was then selected for manual validation. These samples were cross-verified with labels generated by the BirdNet model. The manual validation involved listening to audio recordings and analyzing spectrogram images to match frequencies and other acoustic features. This process revealed that most recordings initially classified as bird sounds were infact insect sounds (primarily cicadas) with significant environmental noise. For species like the White-rumped Shama, BirdNet labels were more accurate compared to the ensemble model's predictions. A secondary comparison of these labels identified six instances of misclassification, (figure 6) highlighting areas for further model improvement. The BirdNet model proved useful for certain avian species, but overall, the findings highlight the importance of a balanced and well-represented dataset and in-domain data for reliable species classification.

5. Conclusion

This study demonstrates that MFCC features and custom CNNs outperform pre-trained models in avian audio classification, especially in identifying bird calls with high precision using labeled data. However, identifying bird species in field recordings using zero-shot learning was impossible due to a lack of domain adaptability, despite using a voting scheme with multiple models. The challenges of species bias and overfitting highlight the need for better model refinement and balanced training datasets. Pre-trained models like YAMNet and Xception performed well in the validation set but failed to generalize to the test set. Manual validation with BirdNet showed high reliability and precision in avian species identification. This approach was the first step towards self-supervised learning, using models trained on other labeled datasets to identify calls and songs from seven specific species in raw, unlabeled field recordings. Future work could use clustering methods to group similar bird sounds and label test data, potentially enhancing accuracy by focusing on nuanced differences in bird sounds using few-shot learning techniques.

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